

4.A2) a) Feed tray is located between stage 6 ✓ 6

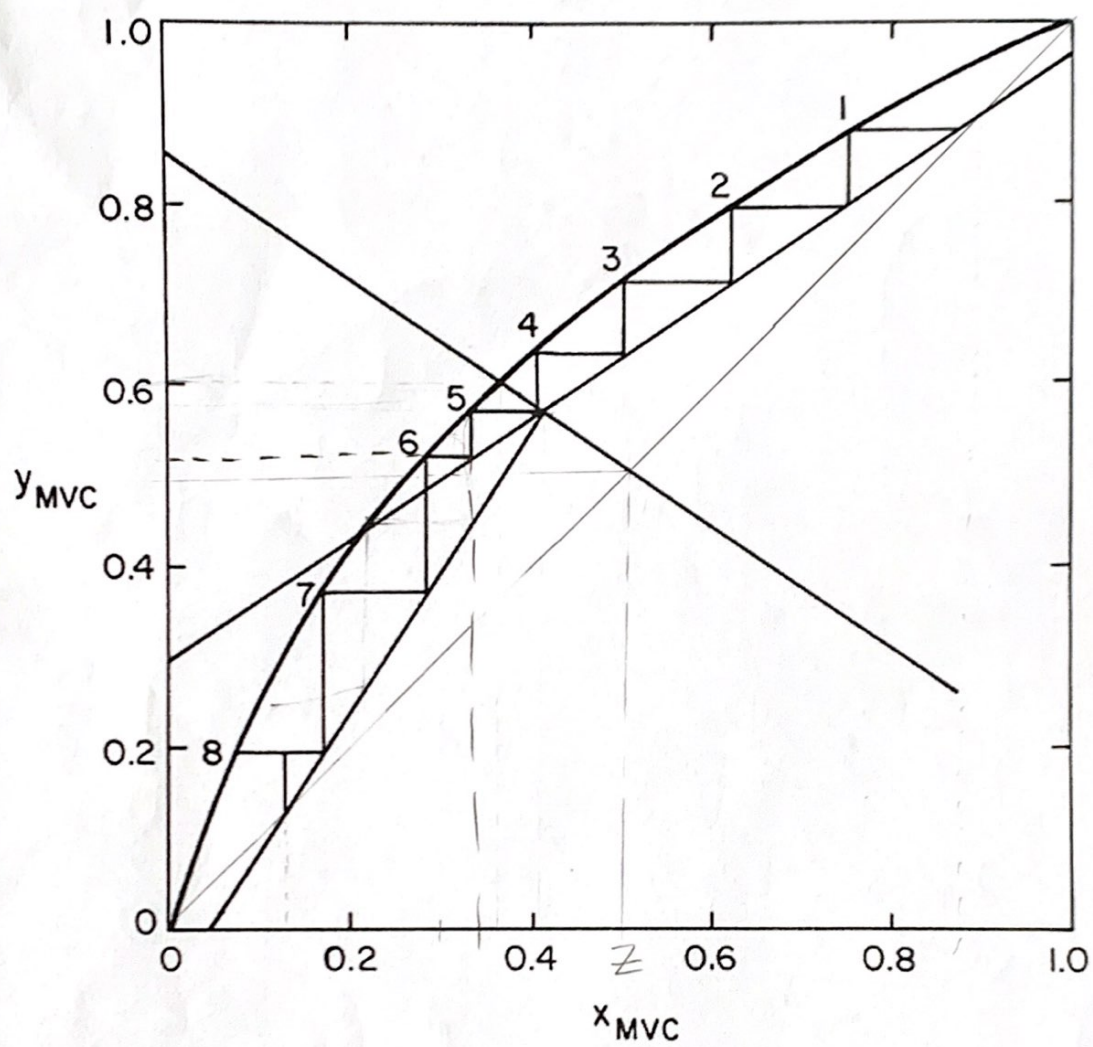
b) $Z = 0.5$

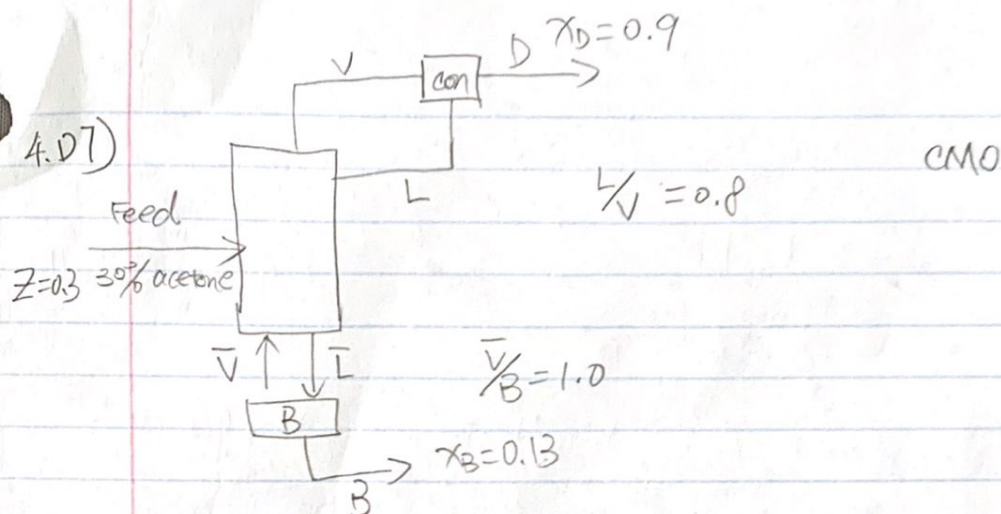
c) $y = 0.52$ stage 6 $\left| \begin{array}{c} \uparrow V_6 \\ \downarrow L_5 \end{array} \right|$

d) $x = 0.33$ ✓

e) two-phase feed b/c the feed has y & x .

f) Temperature at stage 7 is higher than stage 1 ✓





Top op line:

$$y = \frac{L}{V}x + (1 - \frac{L}{V})x_D$$

$$y = 0.8x + 0.2(0.9)$$

$$y = 0.8x + 0.18$$

Bot op line

$$\frac{\bar{V}}{B}=1.0 \Rightarrow \bar{V}=B$$

$$\bar{L} = \bar{V} + B$$

$$\bar{L} = \bar{V} + \bar{V}$$

$$2\bar{V} = \bar{L} \quad \frac{\bar{L}}{\bar{V}} = 2$$

$$y = \frac{\bar{L}}{\bar{V}}x - (\frac{\bar{L}}{\bar{V}} - 1)x_B$$

$$y = 2x - (2 - 1)0.13$$

$$y = 2x - 0.13$$

a) $X_{\text{acetone, 5th}} \approx 0.56$

b) $y_{\text{acetone, 7th}} \approx 0.55$

c) Feed line

$$y = \frac{q}{q-1}x + \frac{1}{1-q}Z$$

passed 2 points

$$(0.3, 0.3)$$

$$(0.26, 0.38)$$

intersection of 2 op lines

$$\text{slope} = \frac{0.3 - 0.38}{0.3 - 0.26} = -2$$

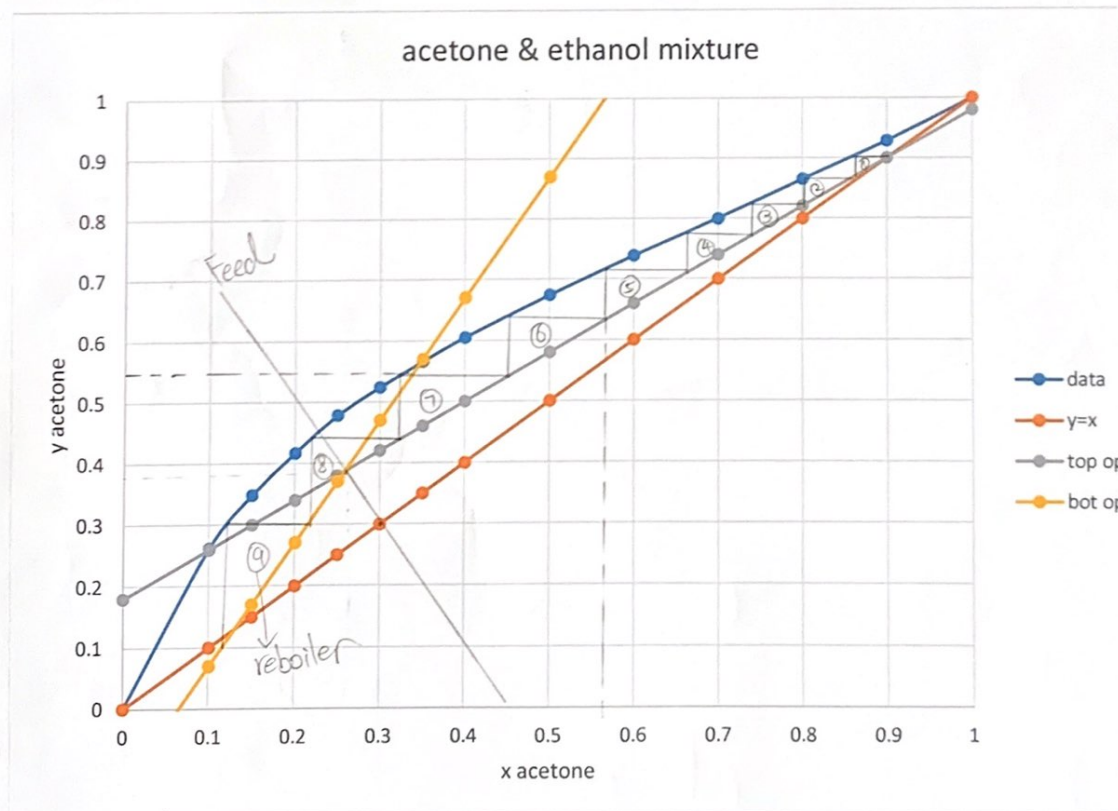
$$\frac{q}{q-1} = -2 \Rightarrow q = \frac{2}{3}$$

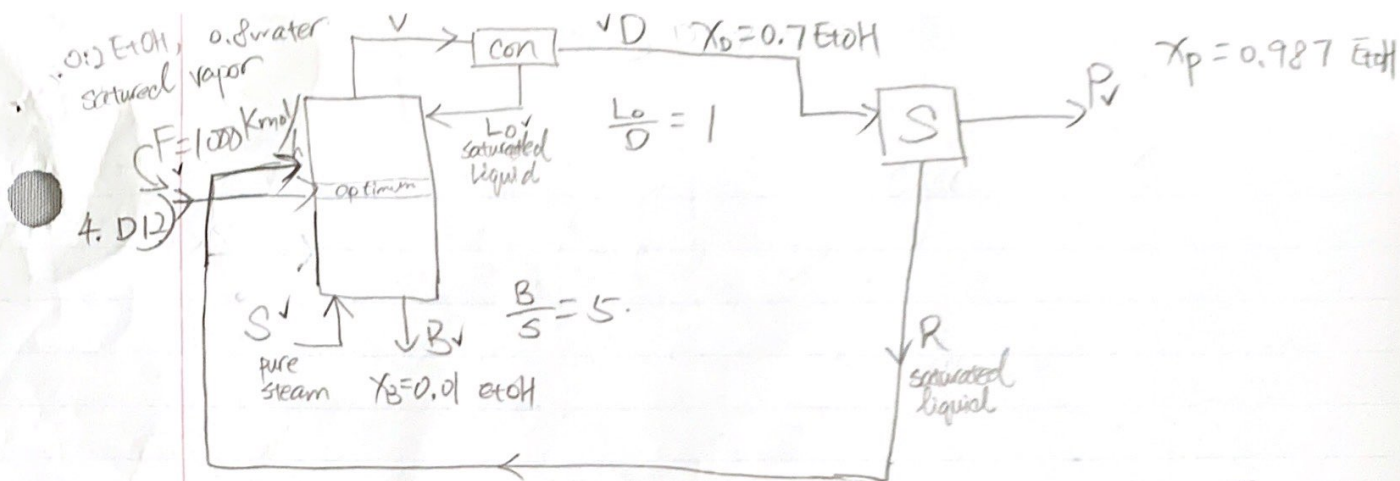
total # of stage: $(9-1) = 8$ stages

optimum stage: stage 8th

reboiler.

x acetone	y acetone	y=x	top op	bot op
0	0	0	0.18	-0.13
0.1	0.262	0.1	0.26	0.07
0.15	0.348	0.15	0.3	0.17
0.2	0.417	0.2	0.34	0.27
0.25	0.478	0.25	0.38	0.37
0.3	0.524	0.3	0.42	0.47
0.35	0.566	0.35	0.46	0.57
0.4	0.605	0.4	0.5	0.67
0.5	0.674	0.5	0.58	0.87
0.6	0.739	0.6	0.66	1.07
0.7	0.8	0.7	0.74	1.27
0.8	0.865	0.8	0.82	1.47
0.9	0.929	0.9	0.9	1.67
1	1	1	0.98	1.87





top

$$V = D + L_0$$

$$\frac{L_0}{D} = 1$$

$$V = L_0 + L_0$$

$$V = 2L_0$$

$$\frac{V}{2} = \frac{L_0}{V}$$

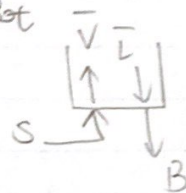
top op line

$$y = \frac{L}{V}x + (1 - \frac{L}{V})X_D$$

$$y = \frac{1}{2}x + \frac{1}{2}(0.7)$$

$$y = \frac{1}{2}x + 0.35$$

Bot



$$B = \bar{L}, \bar{V} = S$$

$$\bar{V} + B = \bar{L} + S$$

$$\bar{V} \cdot y + B \cdot X_B = \bar{L}x + S y^0$$

$$y = \frac{\bar{L}}{\bar{V}}x - \frac{\bar{L}}{\bar{V}}X_B$$

$$\frac{B}{S} = 5$$

$$B = 5S$$

$$\bar{L} = 5\bar{V}$$

$$\frac{\bar{L}}{\bar{V}} = 5 \Rightarrow y = 5x - 5(0.01)$$

$$y = 5x - 0.05$$

$$\bar{F} + S = B + P \quad (B = 5S)$$

$$z \cdot \bar{F} + X_D \cdot S = X_B B + X_P P$$

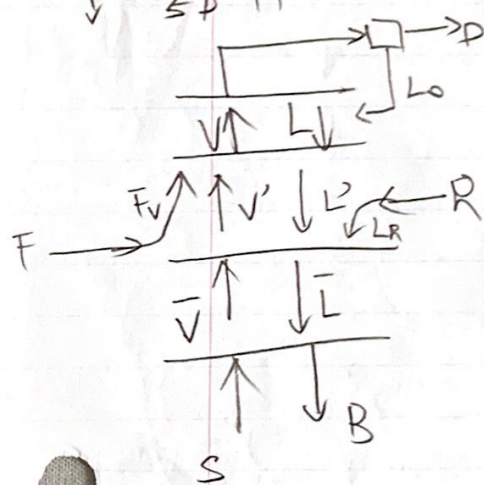
$$0.2 \bar{F} = 0.01B + 0.987P$$

$$\bar{F} = \frac{4}{5}B + P$$

$$\Rightarrow P = 192.408 \text{ kmol/hr}$$

$$B = 1009.49 \text{ kmol/hr}$$

$$\Rightarrow S = \frac{1}{5}B = 201.898$$



$$\bar{V} = S = V'$$

$$F + V' = V \Rightarrow F + S = V$$

$$L + R = \bar{L}$$

$$V = 1201.898 \text{ kmol/hr}$$

$$D = P + R$$

$$R = D - P$$

$$R = 408.541 \text{ kmol/hr}$$

$$D + L_0 = \bar{V}$$

$$2D = V$$

$$D = \frac{1}{2}V = 600.949 \text{ kmol/hr} = L_0$$

$$X_D D = X_R \cdot R + X_P \cdot P$$

$$0.7(600.949) = X_R \cdot (408.541) + 0.987(192.408)$$

$$X_R = 0.5648$$

b) Feed line
 $q=0$

$$y = 0 + \frac{1}{1} z$$

$$y = z$$

$$y = 0.2$$

Recycle line
 $q=1$

$$y = \frac{q}{q-1} x + \frac{1}{1-q} z$$

$$(q-1)y = qx - z$$

$$0 = 1x - z \Rightarrow x = z$$

mid op line

$$y = \frac{L'}{V'} x + \frac{D \cdot X_D - F_R \cdot Z_R}{V'}$$

$$L' = L + R$$

$$= D + R = 600.949 + 408.541$$

$$= 1009.49$$

$$V' = \bar{V} + F$$

$$= S + F$$

$$= 201.898 + 1000$$

$$= 1201.898$$

$$y = \frac{1009.49}{1201.898} x + \frac{600.949 \cdot 0.7 - 408.541 \cdot 0.5648}{1201.898}$$

$$y = 0.8399x + 0.158$$

(stage 1 is in the top)

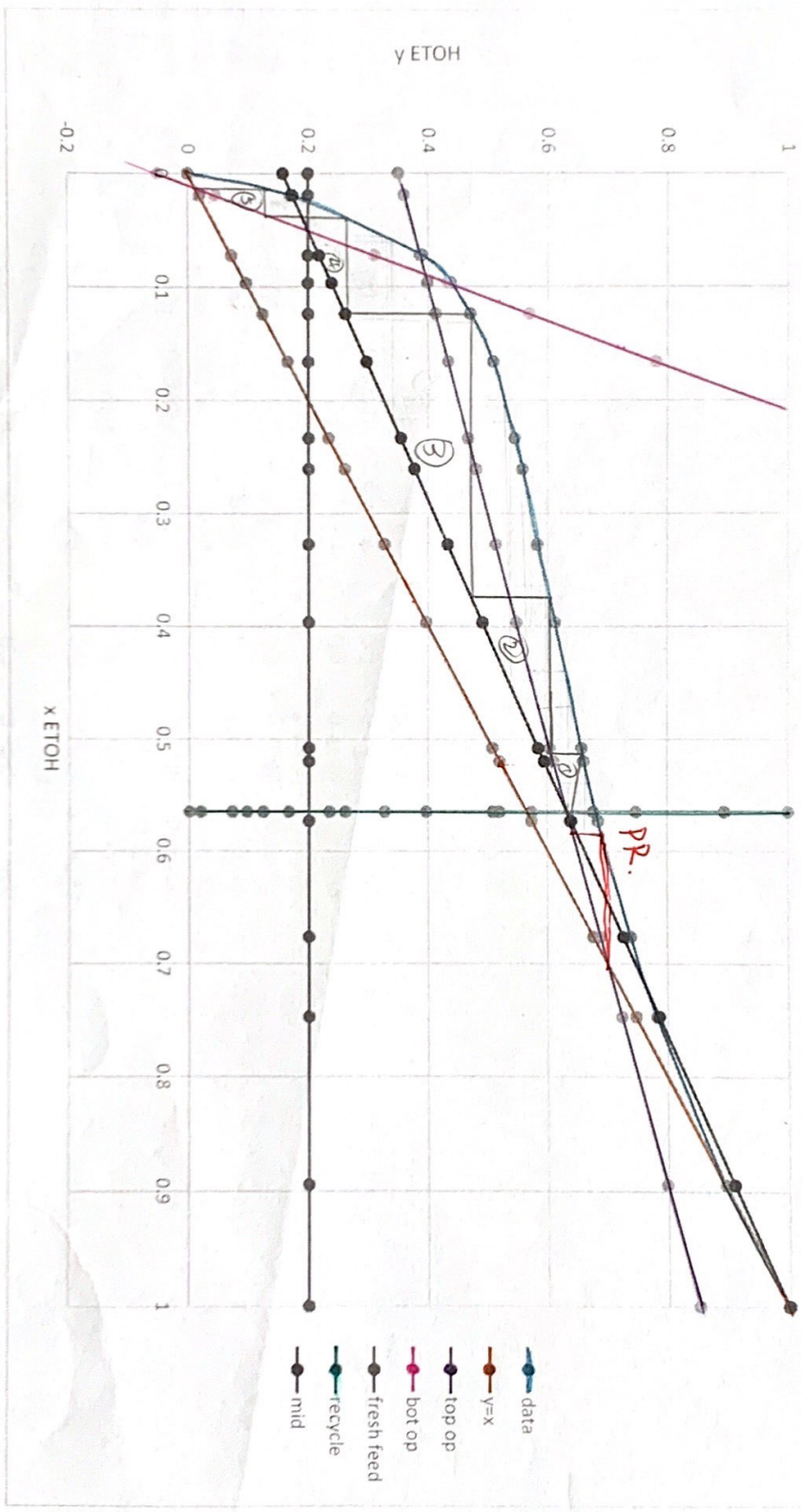
Count the stage from top $X_D = 0.7$

Total # of stages: 5.

optimum feed plate: for fresh feed: stage 4

recycle stream: stage 1

ethanol & water



x EtOH	y EtOH	y=x	top op	bot op	feed line	recycle	mid
0	0	0	0.35	-0.05	0.2	0.5648	0.158
0.019	0.17	0.019	0.3595	0.045	0.2	0.5648	0.173958
0.0721	0.3891	0.0721	0.38605	0.3105	0.2	0.5648	0.218557
0.0966	0.4375	0.0966	0.3983	0.433	0.2	0.5648	0.239134
0.1238	0.4704	0.1238	0.4119	0.569	0.2	0.5648	0.26198
0.1661	0.5089	0.1661	0.43305	0.7805	0.2	0.5648	0.297507
0.2337	0.5445	0.2337	0.46685	1.1185	0.2	0.5648	0.354285
0.2608	0.558	0.2608	0.4804	1.254	0.2	0.5648	0.377046
0.3273	0.5826	0.3273	0.51365	1.5865	0.2	0.5648	0.432899
0.3965	0.6122	0.3965	0.54825	1.9325	0.2	0.5648	0.49102
0.5079	0.6564	0.5079	0.60395	2.4895	0.2	0.5648	0.584585
0.5198	0.6599	0.5198	0.6099	2.549	0.2	0.5648	0.59458
0.5732	0.6841	0.5732	0.6366	2.816	0.2	0.5648	0.639431
0.6763	0.7385	0.6763	0.68815	3.3315	0.2	0.5648	0.726024
0.7472	0.7815	0.7472	0.7236	3.686	0.2	0.5648	0.785573
0.8943	0.8943	0.8943	0.79715	4.4215	0.2	0.5648	0.909123
1	1	1	0.85	4.95	0.2	0.5648	0.9979

